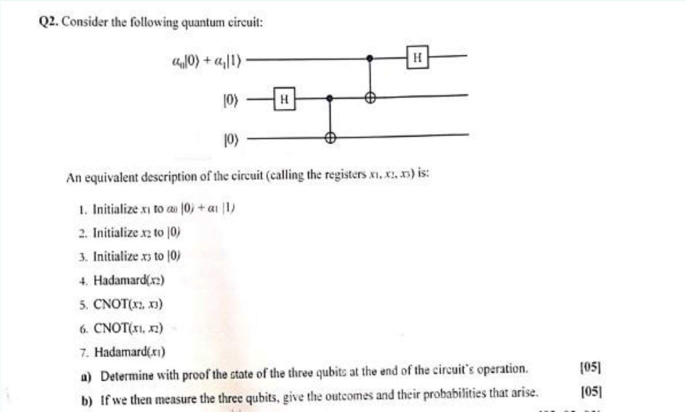
End Sem PYQ



Q2. Consider the following quantum circuit:

An equivalent description of the circuit (calling the registers x_1, x_2, x_3) is:

1. Initialize x_1 to $\alpha_0|0\rangle + \alpha_1|1\rangle$
2. Initialize x_2 to $|0\rangle$
3. Initialize x_3 to $|0\rangle$
4. Hadamard(x_2)
5. CNOT(x_2, x_3)
6. CNOT(x_1, x_2)
7. Hadamard(x_1)

a) Determine with proof the state of the three qubits at the end of the circuit's operation. [05]

b) If we then measure the three qubits, give the outcomes and their probabilities that arise. [05]

answer a

Final 3-qubit state:

$$\frac{1}{2}\Big[a_0(|000\rangle + |011\rangle + |100\rangle + |111\rangle) \; + \; a_1(|010\rangle + |001\rangle - |110\rangle - |101\rangle)\Big]$$

That's the state of (x_1, x_2, x_3) after the circuit.

solve properly

Let x_1, x_2, x_3.

Initial state:

$$(\alpha_0|0\rangle + \alpha_1|1\rangle) \otimes |0\rangle \otimes |0\rangle$$

Step 1: $H(x_2)$

$$|0\rangle \to \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}\big[\alpha_0|0\rangle(|0\rangle + |1\rangle)|0\rangle + \alpha_1|1\rangle(|0\rangle + |1\rangle)|0\rangle\big]$$

Step 2: CNOT($x_2 \to x_3$)

$$= \frac{1}{\sqrt{2}}\left[\alpha_0(|000\rangle + |011\rangle) + \alpha_1(|100\rangle + |111\rangle)\right]$$

Step 3: CNOT($x_1 \to x_2$)

$$= \frac{1}{\sqrt{2}}\left[\alpha_0(|000\rangle + |011\rangle) + \alpha_1(|110\rangle + |101\rangle)\right]$$

Step 4: $H(x_1)$

$$|0\rangle \to \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |1\rangle \to \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Apply:

$$= \frac{1}{2}\Big[\alpha_0(|000\rangle + |100\rangle + |011\rangle + |111\rangle) + \alpha_1(|010\rangle - |110\rangle + |001\rangle - |101\rangle)\Big]$$

Final Answer:

$$\boxed{\frac{1}{2}\Big[\alpha_0(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \alpha_1(|010\rangle + |001\rangle - |110\rangle - |101\rangle)\Big]}$$

b

From part (a), the final state is:

$$\frac{1}{2}\Big[\alpha_0(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \alpha_1(|010\rangle + |001\rangle - |110\rangle - |101\rangle)\Big]$$

Measurement outcomes & probabilities

Each basis state has amplitude:

- $\pm\frac{\alpha_0}{2}$ or $\pm\frac{\alpha_1}{2}$

Probability = (amplitude)^2

Grouping results:

States with α_0:

$$|000\rangle, |011\rangle, |100\rangle, |111\rangle$$

Each has probability:

$$\frac{|\alpha_0|^2}{4}$$

States with α_1:

$$|010\rangle, |001\rangle, |110\rangle, |101\rangle$$

Each has probability:

$$\frac{|\alpha_1|^2}{4}$$

Final Answer (clean):

- $P(000) = P(011) = P(100) = P(111) = \frac{|\alpha_0|^2}{4}$
- $P(010) = P(001) = P(110) = P(101) = \frac{|\alpha_1|^2}{4}$

✔ Total probability = $|\alpha_0|^2 + |\alpha_1|^2 = 1$

Q3. Calculate the following inner products.

a) $\langle 10|11\rangle$.

b) $\langle +-|01\rangle$.

c) $\langle 1+0|1-0\rangle$.

a)

$$\langle 10|11\rangle = 0$$

b)

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$\langle + - |01\rangle = \langle +|0\rangle \cdot \langle -|1\rangle = \frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}$$

c)

$$\langle 1+0|1-0\rangle = \langle 1|1\rangle \cdot \langle +|-\rangle \cdot \langle 0|0\rangle = 1 \cdot 0 \cdot 1 = 0$$

Final Answers:

- a) **0**
- b) **−1/2**
- c) **0**

Q4. (a) If you measure the left qubit, what outcomes can you get, what are the corresponding probabilities of those outcomes, and what does the state collapse to for each outcome? Is this state a product state, partially entangled state, or maximally entangled state? [03+03+03]

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle).$$

Given state:

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

Measurement of left qubit

Possible outcomes: 0 or 1

Outcome: 0

State component: $|01\rangle$

- Probability: $\frac{1}{2}$
- Collapsed state: $|01\rangle$

Outcome: 1

State component: $|10\rangle$

- Probability: $\frac{1}{2}$
- Collapsed state: $|10\rangle$

Nature of state

This is a **maximally entangled state** (Bell state).

(b) Alice and Bob share an EPR-pair,

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Suppose they each measure their qubit with an X-observable (which corresponds to a particular projective measurement with possible outcomes +1, −1). Find the following. **[10]**

(a) P_{00}.

(b) P_{01}.

(c) P_{10}.

(d) P_{11}.

(e) $E(A',B') = P00 - P01 - P10 + P11$.

Given EPR state:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Measuring in **X-basis**:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Rewrite state in X-basis:

$$|00\rangle + |11\rangle = |++\rangle + |--\rangle$$

So,

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$$

Probabilities

- $P_{00} = P(++) = \frac{1}{2}$
- $P_{01} = P(+-) = 0$
- $P_{10} = P(-+) = 0$
- $P_{11} = P(--) = \frac{1}{2}$

Expectation

$$E(A,B) = P_{00} - P_{01} - P_{10} + P_{11} = \frac{1}{2} - 0 - 0 + \frac{1}{2} = 1$$

Final Answers:

- $P_{00} = \frac{1}{2}$
- $P_{01} = 0$

- $P_{10} = 0$
- $P_{11} = \frac{1}{2}$
- $E(A, B) = 1$

(iii) NAND (iv) NOR

Q7. (a) Let's assume:

Alice's key:

A = 1 0 1 1 0 1 1 0

Bob's key:

B = 1 0 1 1 1 1 1 0

The above Alice and Bob raw keys contain discrepancies. Demonstrate how the Cascade protocol can be used to detect and correct errors. [5]

Given:

- Alice: **10110110**
- Bob: **10111110**

Step 1: Divide into blocks (size = 4)

- Block 1:
  A = 1011
  B = 1011 → parity same → **no error**
- Block 2:
  A = 0110
  B = 1110 → parity different → **error present**

Step 2: Binary search in Block 2

Split into 2 parts:

Part 1:

- A = 01
- B = 11 → parity mismatch → error here

Split further:

- A = 0, B = 1 → ❌ error found → **correct Bob's bit to 0**

Now Block 2 becomes: **0110**

Step 3: Final corrected key

Bob's corrected key:

$$\boxed{10110110}$$

Final Answer:

- Error detected at position **5**
- Corrected key: **10110110**
- Keys now match ✔️

Q6. (a) Explain the Crystal-Kyber public encryption algorithm. [6]

Explain Crystal Kyber in simple steps to write in exam

Here's a **clean, exam-ready answer** you can memorize and write 👇

Crystal-Kyber (CRYSTALS-Kyber) – Simple Explanation

Definition:
CRYSTALS-Kyber is a **post-quantum public key encryption algorithm** based on the **Module Learning With Errors (MLWE)** problem. It is secure against quantum attacks.

---

1. Key Generation

1. Generate a random matrix A
2. Choose small secret vector s
3. Generate small error vector e
4. Compute:

$$t = A \cdot s + e$$

- **Public Key:** (A, t)
- **Private Key:** s

---

2. Encryption

1. Convert message m into vector form

2. Choose random vector r and errors e_1, e_2
3. Compute:

$$u = A^T \cdot r + e_1$$

$$v = t^T \cdot r + e_2 + m$$

- **Ciphertext:** (u, v)

---

3. Decryption

1. Use private key s
2. Compute:

$$v - s^T \cdot u$$

3. Remove noise → recover message m

---

4. Key Idea

- Security comes from **Lattice problems (MLWE)**
- Noise makes it hard for attackers to recover s

---

5. Advantages

- Quantum-resistant
- Fast and efficient
- Small key sizes
- Selected by **NIST** for post-quantum cryptography

Final Line (important for exam)

👉 *CRYSTALS-Kyber is a lattice-based, quantum-secure public key encryption scheme that uses the MLWE problem and noise to ensure security.*

2/2

similarly explain CRYSTALS-Dilithium

Here's a **clean, exam-ready answer** 👇

CRYSTALS-Dilithium – Simple Explanation

Definition:
CRYSTALS-Dilithium is a **post-quantum digital signature algorithm** based on the **Module Learning With Errors (MLWE)** problem. It is used for **authentication and integrity** and is secure against quantum attacks.

---

1. Key Generation

1. Generate matrix A
2. Choose small secret vectors s_1, s_2
3. Compute:

$$t = A \cdot s_1 + s_2$$

- **Public Key:** (A, t)
- **Private Key:** (s_1, s_2)

---

2. Signing

1. Take message m
2. Generate random vector y
3. Compute:

$$w = A \cdot y$$

4. Generate challenge $c = H(w, m)$
5. Compute:

$$z = y + c \cdot s_1$$

- **Signature:** (z, c)

---

3. Verification

1. Compute:

$$w' = A \cdot z - c \cdot t$$

2. Check:

$$c \stackrel{?}{=} H(w', m)$$

- If equal → **valid signature**
- Else → **reject**

---

4. Key Idea

- Based on **lattice problems (MLWE)**
- Uses hashing + randomness for security
- Prevents forgery even with quantum computers

---

5. Advantages

- Quantum-resistant
- Efficient and fast
- Strong security guarantees
- Selected by **NIST** for digital signatures

Final Line (write this for marks)

👉 *CRYSTALS-Dilithium is a lattice-based, quantum-secure digital signature scheme that uses MLWE and hashing to provide authentication and integrity.*

Q1. (a) Explain Shor's Algorithm and describe through a suitable example how it can pose a threat to RSA. **[6]**

Here's a **clean, exam-ready answer** you can directly write 👇

Shor's Algorithm – Explanation

Definition:
Shor's Algorithm is a **quantum algorithm used to factor large integers efficiently**, which is hard for classical computers.

Key Idea

- It converts **factorization** into a **period-finding problem**
- Uses **quantum parallelism + Quantum Fourier Transform (QFT)** to find the period quickly

---

Steps of Shor's Algorithm

1. Choose a random number $a < N$ such that $\gcd(a, N) = 1$
2. Define function:

$$f(x) = a^x \mod N$$

3. Use a quantum computer to find the **period** r of $f(x)$
4. If r is even, compute:

$$\gcd(a^{r/2} - 1, N), \quad \gcd(a^{r/2} + 1, N)$$

5. These give **non-trivial factors of** N

Example (Simple)

Let $N = 15$

1. Choose $a = 2$
2. Compute:

$$2^x \mod 15 = 2, 4, 8, 1, \ldots$$

3. Period $r = 4$
4. Compute:

$$\gcd(2^2 - 1, 15) = \gcd(3, 15) = 3$$

$$\gcd(2^2 + 1, 15) = \gcd(5, 15) = 5$$

👉 Factors: **3 and 5**

Why it Threatens RSA

- RSA security depends on **difficulty of factoring large numbers**
- Classical computers take **exponential time**
- Shor's algorithm does it in **polynomial time**

👉 So, a powerful quantum computer can **break RSA keys easily**

Final Line (important)

👉 *Shor's algorithm efficiently factors large integers using quantum period finding, making classical cryptosystems like RSA insecure in the presence of quantum computers.*